NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Centre for Quantum Technologies

Transfer of Skyrmion structure from Light to Ultracold Atomic Ensemble

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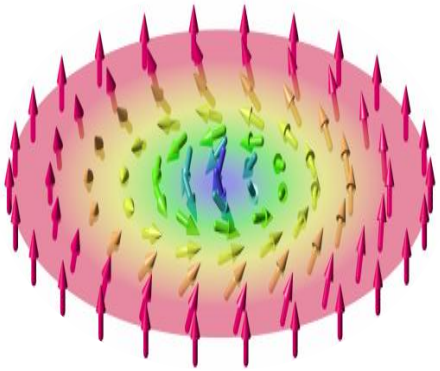
CLEO Pacific Rim

August 5 2024

Skyrmions

- Magnetic Skyrmions 2D magnetization textures in magnetic materials

  → spin configurations with non-trivial topology
  ⇒ cannot be smoothly deformed to the magnetic ground state
  → localized



Theoretical proposal: A. N. Bogdanov and D. A. Yablonskii, Zh. Eksp. Teor. Fiz. 95,178-182 (1989)
First experimental observation: S. Mühlbauer *et al.*, *Science* **323**, 915-919 (2009)

↪ spintronics, magnetic storage/computation

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Skyrmions

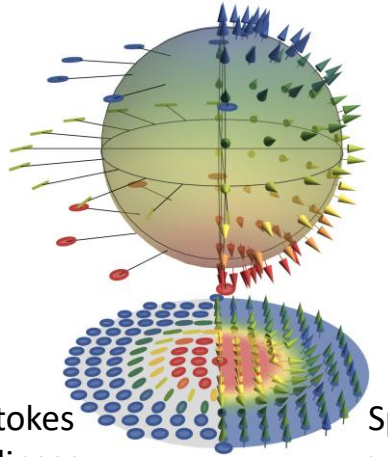
- <u>Optical Skyrmions</u>
  → Surface plasmon polaritons (evanescent waves) S. Tsesses *et al.*, *Science* 361, 993-996 (2018)
  → Full Poincaré beams (free space) S. Gao *et al.*, Phys. Rev. A 102, 053513 (2020)

↪ study of topological photonics, optical communication

Bloch or Poincaré sphere

Stereographic projection

Real space



Stokes ellipses

Spin vectors

Skyrmion winding number:

$$n = \frac{1}{4\pi} \iint \vec{S} \cdot (\partial_y \vec{S} \times \partial_x \vec{S}) dx dy$$

$\vec{S}$ represents the spin, magnetization or Stokes vector

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Outline

- Production and characterization of the optical Skyrmion
- Transfer and characterization of the atomic Skyrmion spin texture

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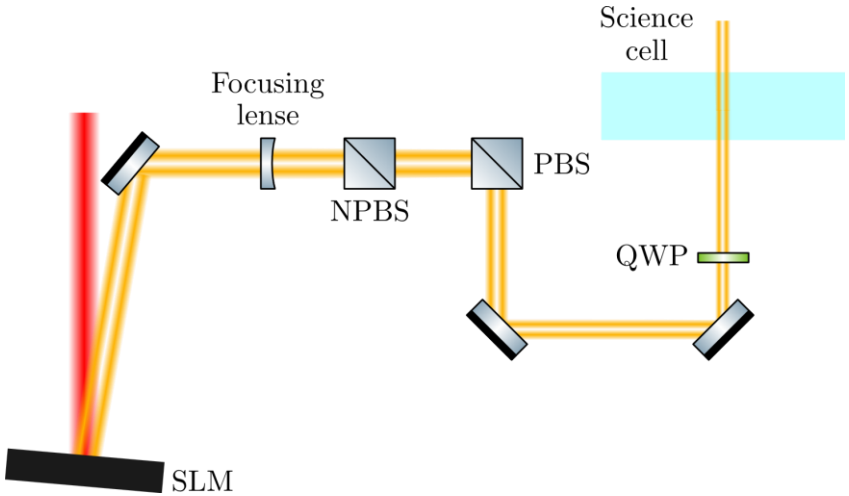
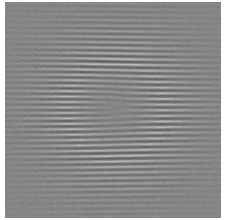
Creating the optical Skyrmion

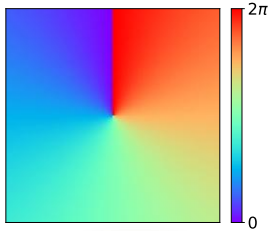




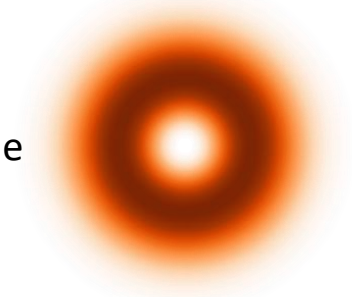
Phase mask obtained using a technique from: Arrizón *et al.*, J. Opt. Soc. Am. A 24, 11 (2007)

Laguerre-Gaussian beam
Radial index $p = 0$
Azimuthal index $l = 1$

Phase profile



Intensity profile



Right-handed circular polarization

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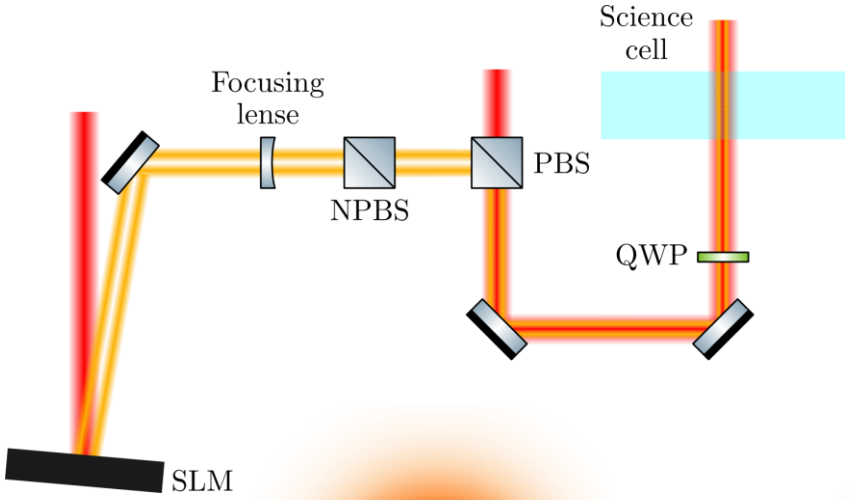
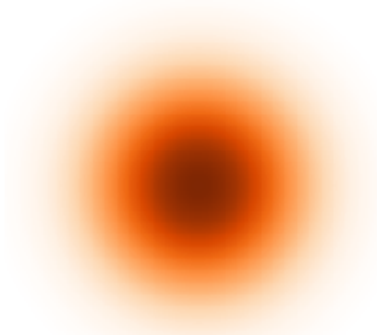
Creating the optical Skyrmion



Overlapping a Gaussian beam with left-handed polarization creates the optical Skyrmion



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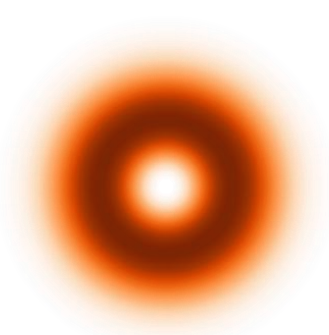

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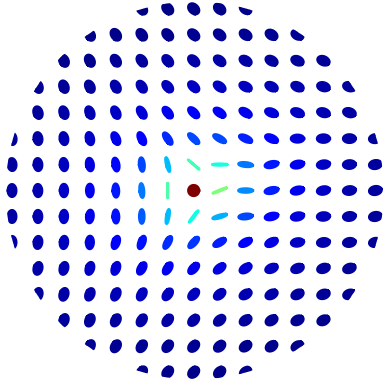

Polarization state $\cos\frac{\theta_o}{2}|L\rangle + e^{i\phi_o}\sin\frac{\theta_o}{2}|R\rangle$

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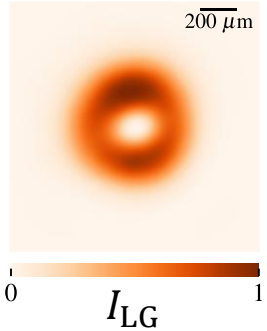
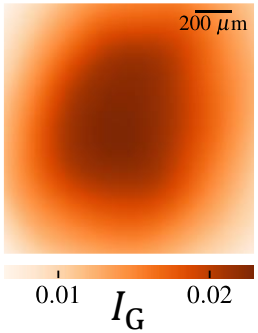
Optical Skyrmion characterization

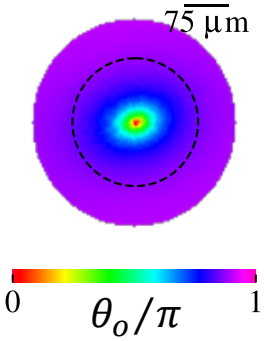
Intensity profiles (10 averages)

Mixing angle θ_o



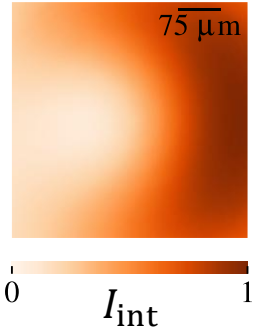


$$\tan\frac{\theta_o}{2} = \sqrt{\frac{I_{LG}}{I_G}}$$

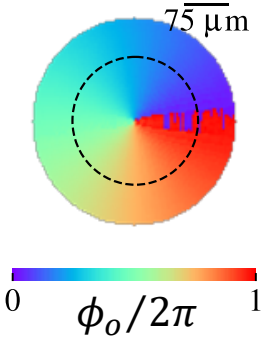


Interference (100 averages)

Phase angle ϕ_o



$$I_{int} = A + B\cos\phi_o$$



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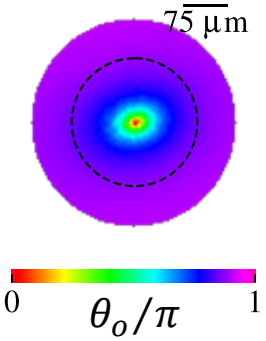
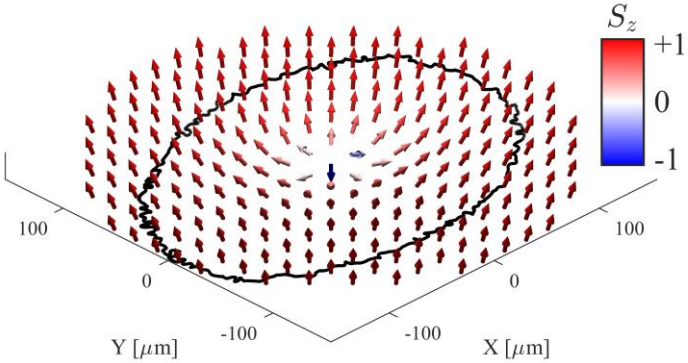
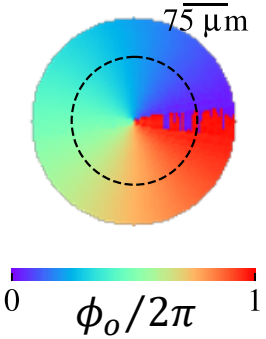
Optical Skyrmion characterization



$$\cos\frac{\theta_o}{2}|L\rangle + e^{i\phi_o}\sin\frac{\theta_o}{2}|R\rangle$$



Poincaré sphere



Skyrmion winding number:
$n_o \approx 0.92$

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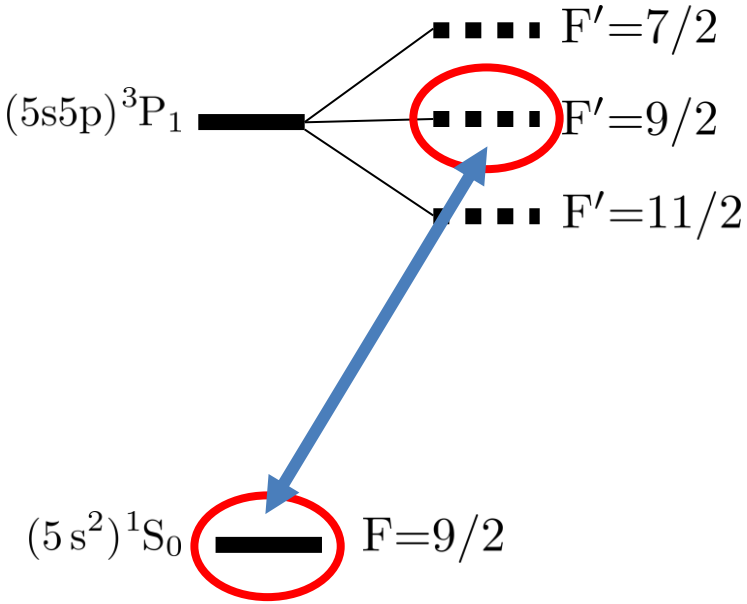
Strontium-87

Isotope	Abundance %
^{88}Sr	82.58
^{86}Sr	9.86
^{87}Sr	7.00
^{84}Sr	0.56

Nuclear spin = 9/2



Ultracold atomic ensemble:

4×10^6 atoms
$7\ \mu K$
107 μm cloud size

Intercombination line at 689 nm
Linewidth $\Gamma = 7$ kHz

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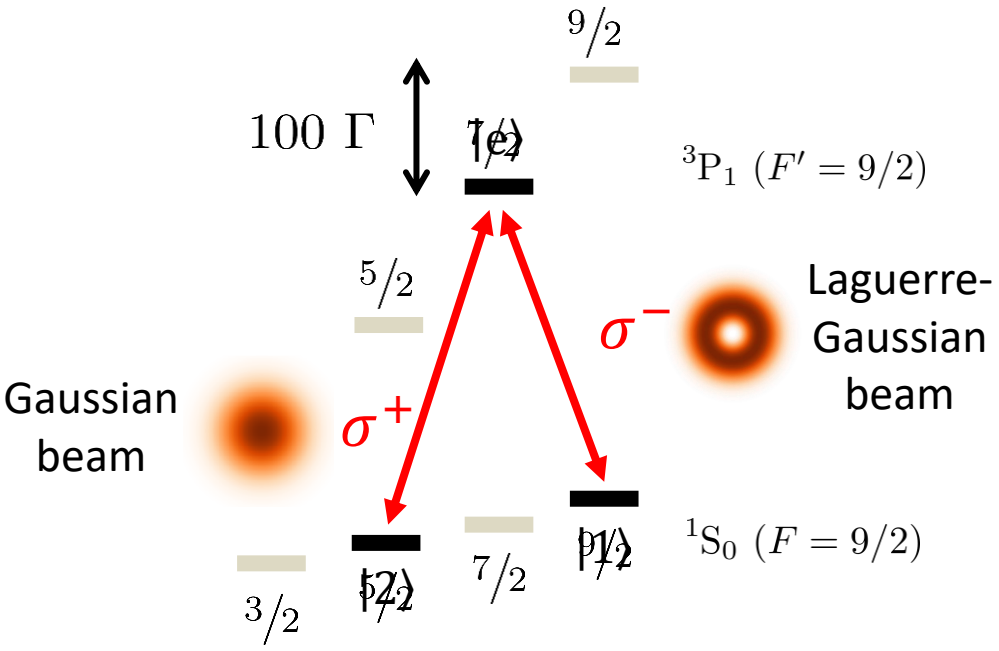
Skyrmion transfer to atoms



In this work: pseudo spin-1/2 system using ground state manifold

10 Gauss bias magnetic field

Zeeman shift → isolated lambda system

'Dark' state:

$$|D\rangle = \Omega_2|1\rangle - \Omega_1|2\rangle$$

Rabi frequencies $|\Omega_i| \propto \sqrt{I_i}$

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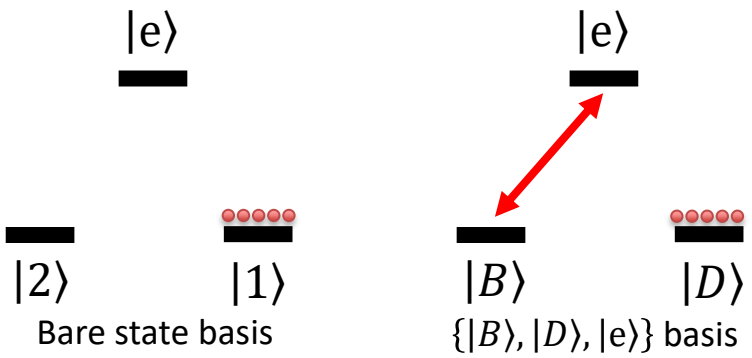
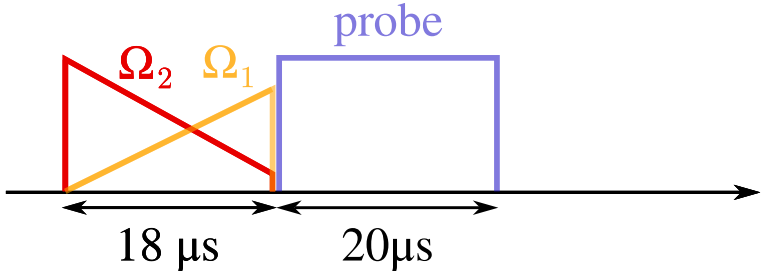
Atomic Skyrmion characterization

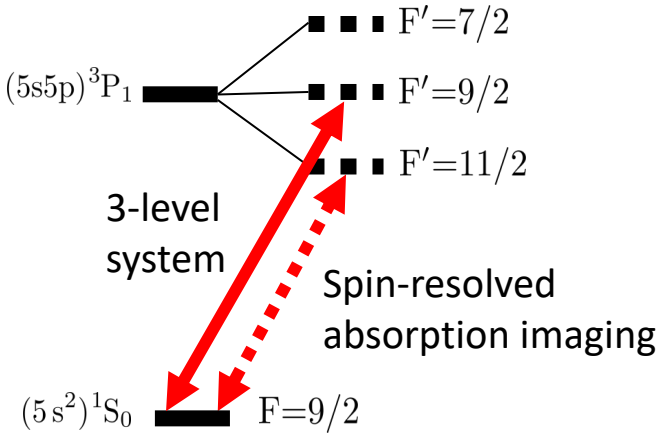
Population measurements



$$|D\rangle = \Omega_2|1\rangle - \Omega_1|2\rangle$$
$$|D\rangle = \cos\frac{\theta_a}{2}|1\rangle + e^{i\phi_a}\sin\frac{\theta_a}{2}|2\rangle$$



- Start with atoms optically pumped in the ground state $|1\rangle$
- Ramp the laser powers **slowly** so that atoms remain in the dark state (adiabaticity)
- Measure population using absorption imaging



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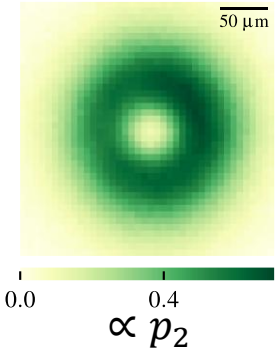
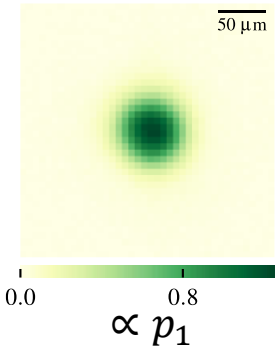
Atomic Skyrmion characterization

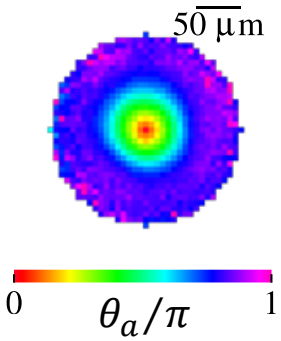
Optical densities (143 averages)

Mixing angle θ_a





$$\tan\frac{\theta_a}{2} = \sqrt{\frac{p_2}{p_1}}$$



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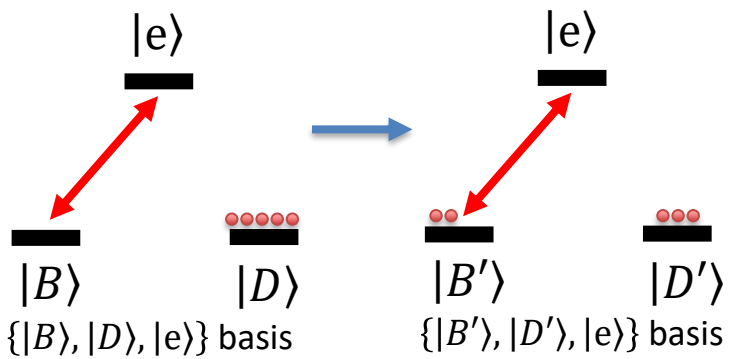
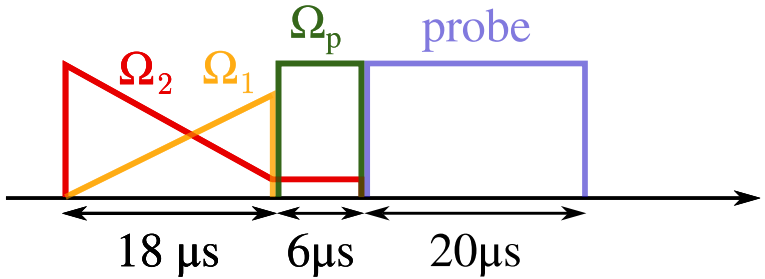
Atomic Skyrmion characterization

Phase measurement



$$|D\rangle = \Omega_2|1\rangle - \Omega_1|2\rangle$$
$$|D\rangle = \cos\frac{\theta_a}{2}|1\rangle + e^{i\phi_a}\sin\frac{\theta_a}{2}|2\rangle$$



- Abrupt (non-adiabatic) switch to a Gaussian beam between $|1\rangle \leftrightarrow |e\rangle$

$|D\rangle$ projects on new basis:

$$|B'\rangle = \Omega_2|1\rangle + \Omega_p|2\rangle$$
$$|D'\rangle = \Omega_2|1\rangle - \Omega_p|2\rangle$$

- Atoms get coupled to the excited state (depletion in ground state population)

- Measure population in state $|2\rangle$ using absorption imaging after π-pulse

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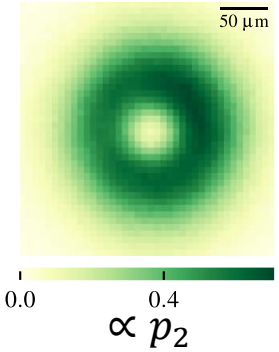
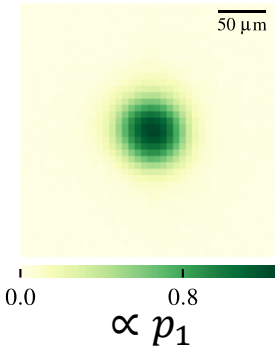
Atomic Skyrmion characterization

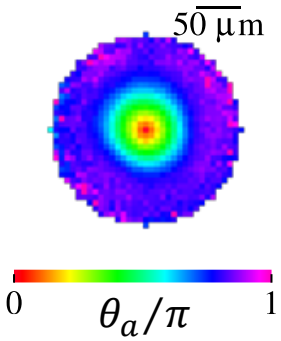
Optical densities (143 averages)

Mixing angle θ_a



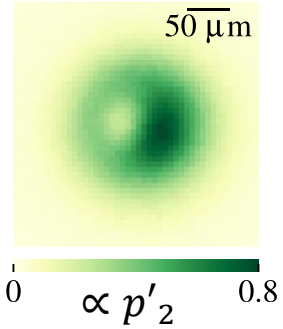


$$\tan\frac{\theta_a}{2} = \sqrt{\frac{p_2}{p_1}}$$

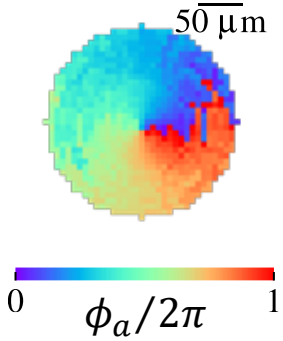


Interference (143 averages)

Phase angle ϕ_a



$$p'_2 = A + B\cos\phi_a$$



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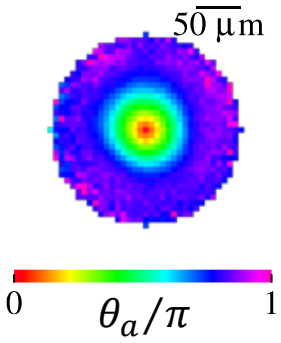
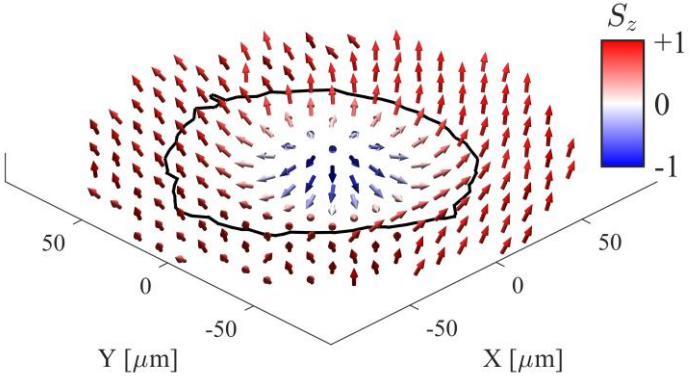
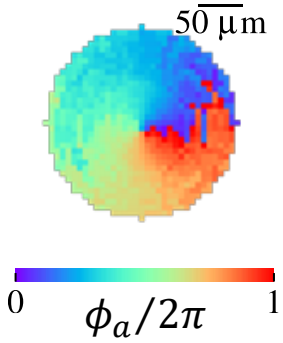
Atomic Skyrmion characterization



$$\cos\frac{\theta_a}{2}|1\rangle + e^{i\phi_a}\sin\frac{\theta_a}{2}|2\rangle$$



Bloch sphere



Skyrmion winding number:
$n_a \approx 0.86$

By taking the same θ value for the contour the optical case gives a winding number of 0.85

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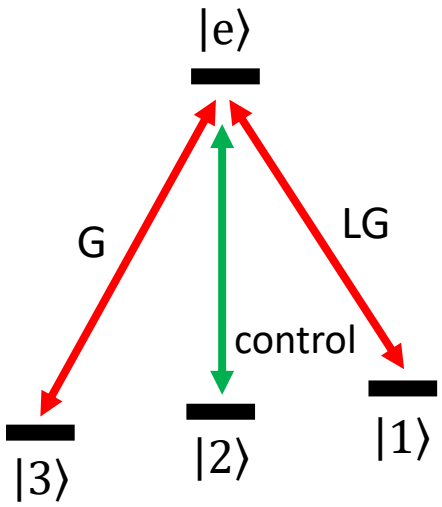
Summary and Outlook

- Coherent excitation of an atomic gas using an optical Skyrmion beam
- Information about the topology is transferred to the atoms



- Study the excitation of atomic ensembles with light having more complicated topologies
- Storage and retrieval of an optical Skyrmion in a four-level system

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Thank you!

Experimental team

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(PhD)

Chetan Sriram Madasu
(postdoc)

Chang Chi Kwong
(senior postdoc)

David Wilkowski
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Theory collaborators

Asst Prof Yijie Shen
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